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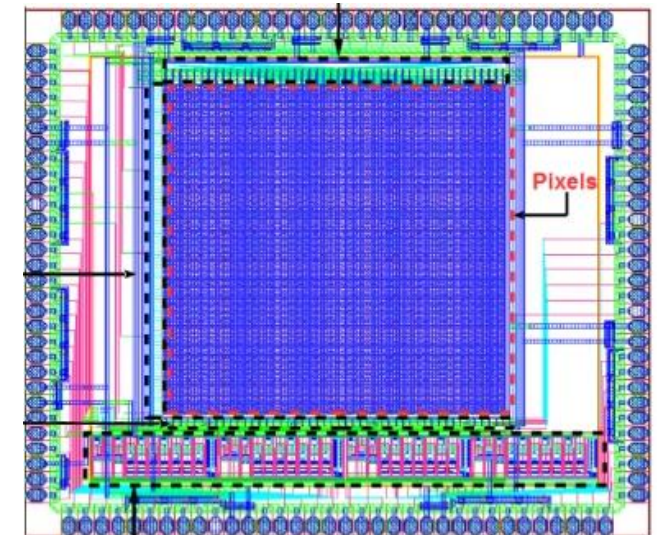
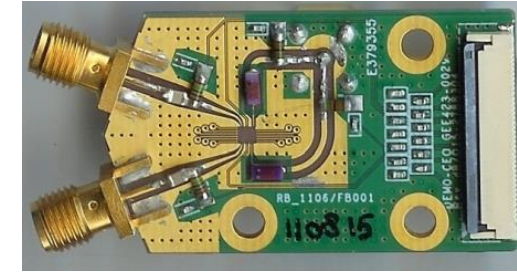
Electronic Systems Design and Automation

S4 EAC

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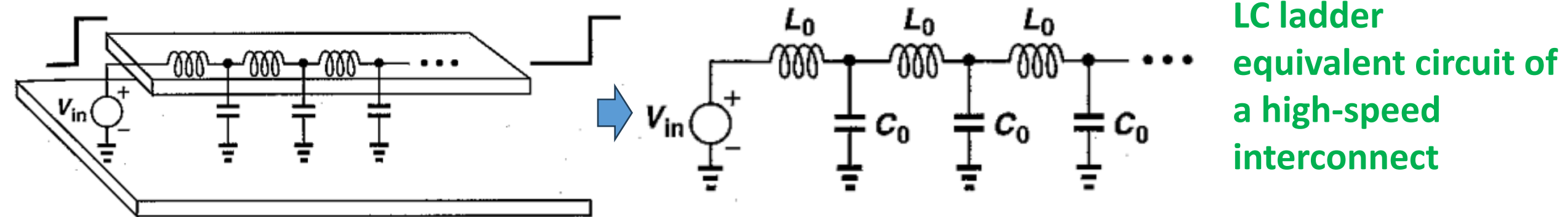
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High-speed PCB design



The trace as a Transmission line

- A transmission line (T line) is an interconnect whose length is a significant fraction of the wavelength of interest
- Its end-to-end delay is not negligible with respect to other time scales like rise time and pulse period.

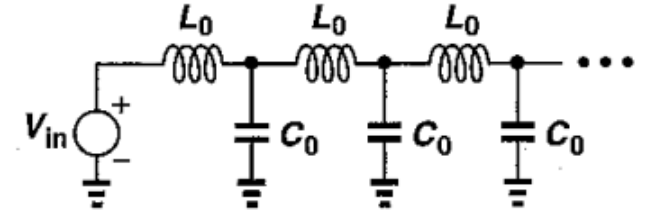


- Such an ideal interconnect can be modelled using series inductance and shunt capacitor distributed along the length of the interconnect – LC ladder
- L_0 -Inductance per unit length
- C_0 -Capacitance per unit length

Characteristic impedance Z_0

- The velocity of propagation in such an interconnect can be written as

$$v = \frac{1}{\sqrt{L_0 C_0}} = \frac{1}{\sqrt{\epsilon \mu}} = \frac{c}{\sqrt{\epsilon_r}}$$

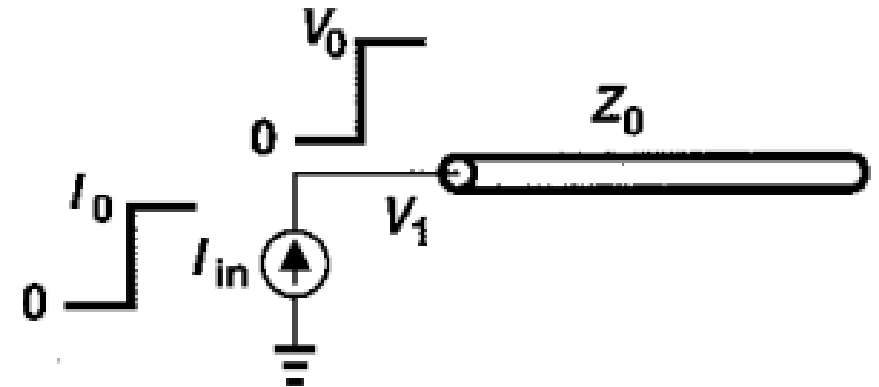


- The propagation delay in T lines has an interesting property
- If a step is applied to one end, the instantaneous behavior at that end does not depend on the length of the line or the load seen at the other end
- An input current step gives rise to a voltage step whose height is a fundamental property of the line called characteristic impedance

$$V_0 = I_0 Z_0$$

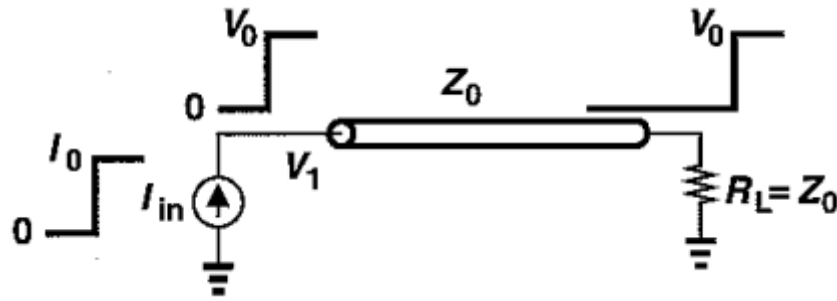
- Where, the characteristic impedance of the line is

$$Z_0 = \sqrt{\frac{L_0}{C_0}}$$



Impedance matched $Z_L = Z_0$

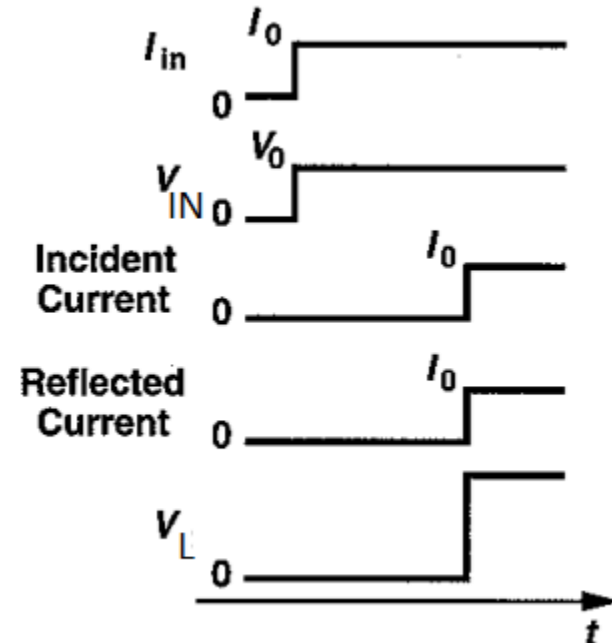
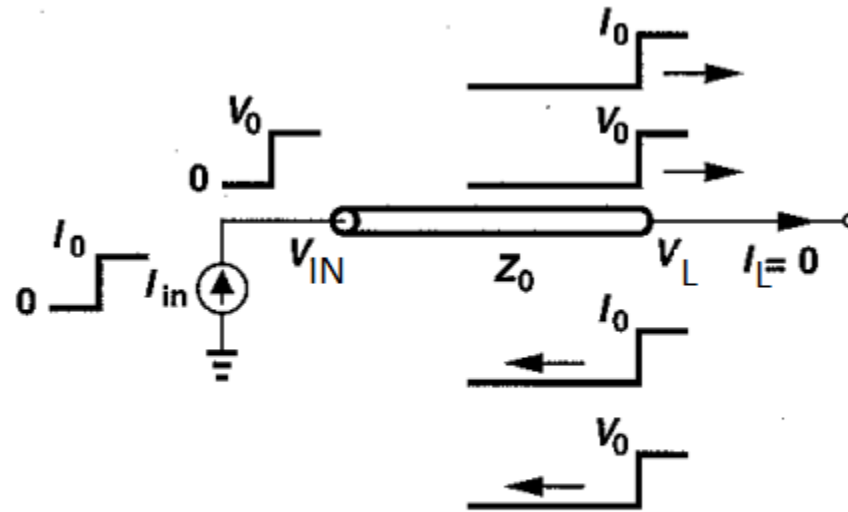
- $Z_0 = \sqrt{\frac{L_0}{C_0}}$ is a real number for lossless traces
- So characteristic impedance is that impedance seen instantaneously by the input



- The voltage pulse at the input further propagates to the end of the trace
- At each point, the current is I_0 and the voltage is V_0 if the load connected at the far end is also Z_0
- The transient is absorbed by the load, and nothing comes back
- This condition is called impedance matched case
- All the power is delivered to the load.

Impedance mismatched $R_L \neq Z_0$

- If $R_L \neq Z_0$, then there should be some reflection since the resistance there is different and the voltage and current will have to be different from V_0 and V_0/Z_0
- Together, the incident pulse and reflected pulse when added gives the required relation at the load ie. $R_L = \frac{V_L}{I_L} = \frac{V_0 + V_{reflected}}{I_0 + I_{reflected}}$ (in a simplified relation)
- If it is an open circuit $R_L = \infty$ then $V_{reflected} = V_0$ and $I_{reflected} = -I_0$ giving $I_L = 0$ (no current flows into an open circuit)
- If it is a short circuit $R_L = 0$ then $V_{reflected} = -V_0$ and $I_{reflected} = I_0$ giving $V_L = 0$ (no voltage across a short circuit)



Input Impedance when mismatched Z_{IN}

- The input impedance with a mismatched load R_L will then also be different from Z_0

$$Z_{IN} = \frac{V_{IN}}{I_{IN}}$$

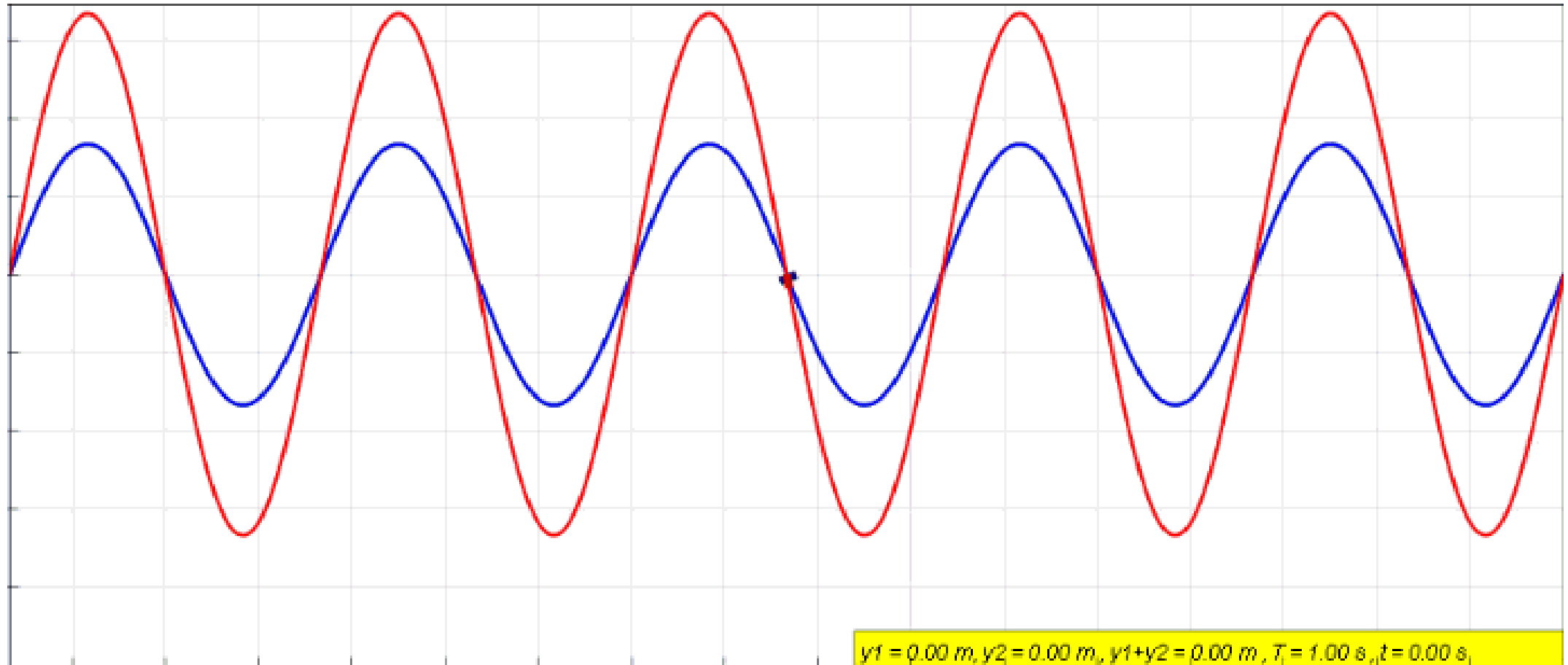
- The above expressions, while considering the length of the line l , will be

$$Z_{IN} = Z_0 \frac{Z_L + jZ_0 \tan \beta l}{Z_0 + jZ_L \tan \beta l}$$

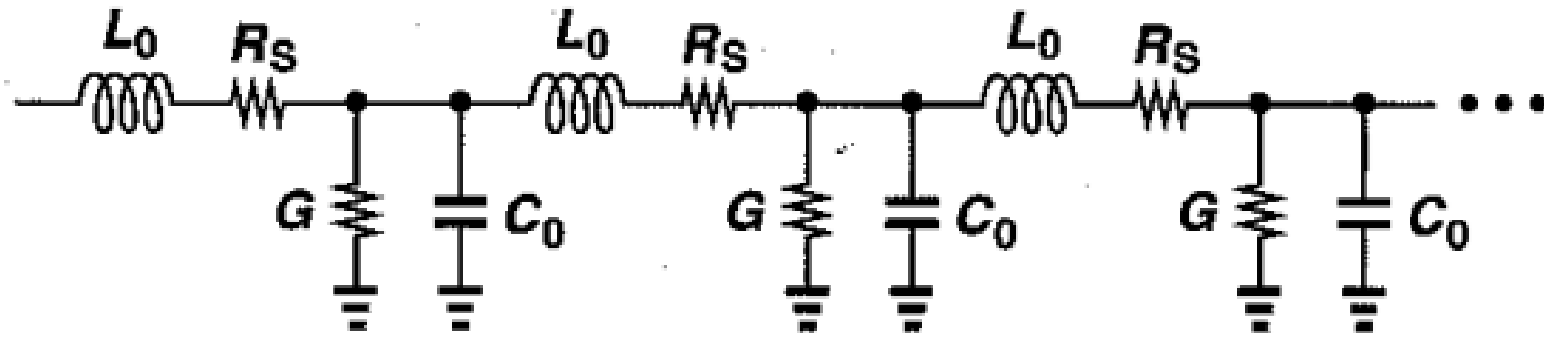
ϵ is the permittivity
 μ is the permeability
 λ_g is the wavelength of the wave in the trace

- Here β is the phase constant given by $\beta = \frac{2\pi}{\lambda_g} = \omega\sqrt{LC} = \omega\sqrt{\epsilon\mu}$
- βl is called **electrical length** in degrees or radians (phase acquired over length l)
- If length is **quarter** of a wavelength $Z_{IN} = \frac{Z_0^2}{Z_L}$ (i/p impedance is proportional to $1/Z_L$)
- If length is **half** of a wavelength $Z_{IN} = Z_L$
- If load is an **open circuit**, then $Z_{IN} = -jR_0 \cot \beta l$ (-ve reactance $\equiv$ a capacitor for short l)
- If load is a **short circuit**, then $Z_{IN} = jR_0 \tan \beta l$ (+ve reactance $\equiv$ inductor for short l)

- If you send sine waves, then you will get standing waves

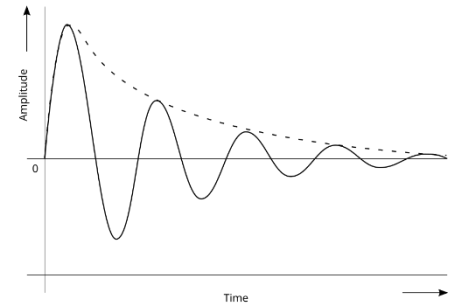


If the line has losses



- If the line has losses, then
- LCRG ladder will be used
- There will be a drop in the voltage and current
- Thus, the impedance will include a series resistance R and shunt admittance G per unit length

$$Z_0 = \sqrt{\frac{R + j\omega L}{G + j\omega C}}$$



- The propagation constant $\gamma = \alpha + j\beta = \sqrt{(R + j\omega L)(G + j\omega C)}$
- The propagating voltage will be decreasing due to an attenuation coefficient α
- There will be phase change due to a phase constant β
- The forward propagating voltage goes as a decreasing wave
$$V(z) = V_0 e^{-\alpha z} \cos(\omega t - \beta z)$$

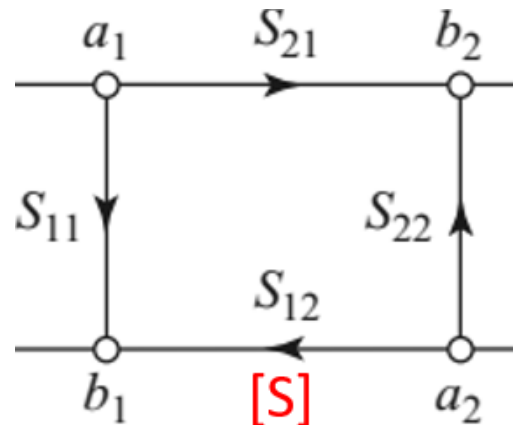
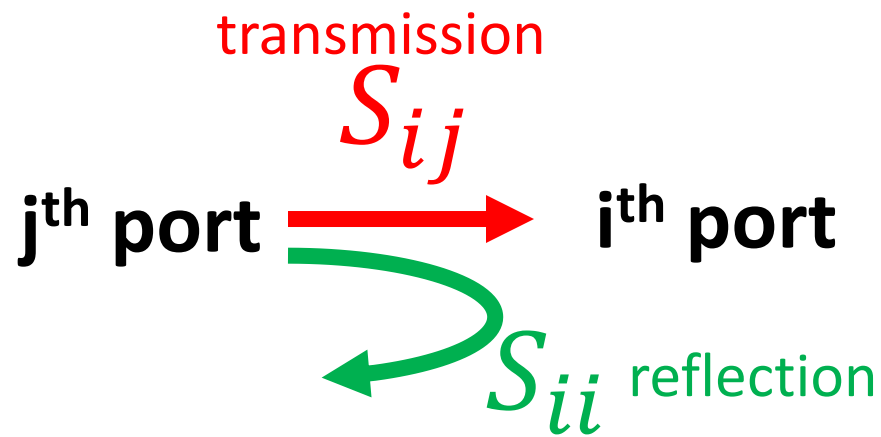
$$\beta \cong \omega \sqrt{LC}$$
$$\alpha \cong \frac{1}{2} \left(R \sqrt{\frac{C}{L}} + G \sqrt{\frac{L}{C}} \right)$$

Reflection and transmission

- The ratio of the reflected wave amplitude to the incident is called reflection coefficient Γ and can be calculated as $\Gamma = \frac{Z_L - Z_0}{Z_L + Z_0}$
- What is not reflected goes through to the load and is called transmission coefficient $\tau = 1 + \Gamma$
- Since sum of reflected powers are proportional to squares of voltages, $P_r = P_i |\Gamma|^2$
- The reflection terms, when expressed in -ve of dB are called return losses
$$RL = -20 \log |\Gamma| \text{ [dB]}$$
- For a matched load ($\Gamma = 0$), the return loss is ∞ .
- For total reflection ($\Gamma = 1$), the return loss is 0.
- The transmission terms when expressed in -ve of dB are called Insertion losses
$$IL = -20 \log |\tau| \text{ [dB]}$$
- 100% transmission => IL=0 dB
- 0% transmission => IL= ∞

Scattering parameters

- The reflection coefficient of a device is often represented by a scattering parameter S_{11} , S_{22} , S_{33} etc. i.e. S_{ii} where the repeated index indicates the reflected (output) wave and the second index indicates the incident (input) wave are both in the same port (like a terminal)
- These scattering parameters are measured while exciting the port with the second index while keeping all other ports in matched condition
- The transmission coefficient of a device to any port is often represented by a scattering parameter S_{21} , S_{12} , S_{13} , etc., i.e. S_{ij} where the first index indicates the reflected (output) wave and the second index indicates the incident (input) wave
- A circuit with N ports will have N reflection terms and $N^2 - N$ transmission terms

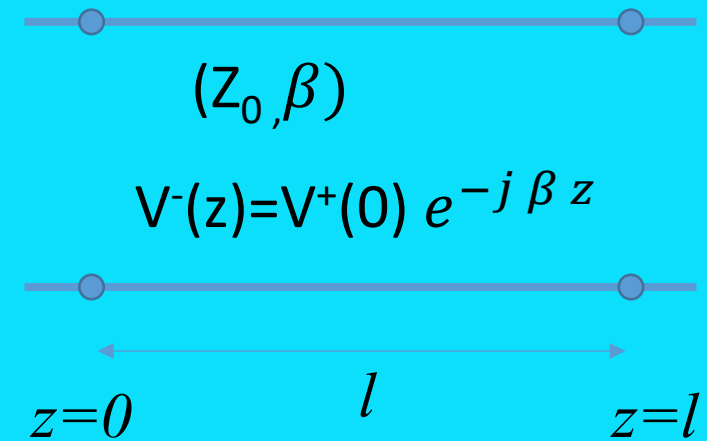


$$S_{ij} = \left. \frac{b_i}{a_j} \right|_{a_k=0 \text{ for } k \neq j}$$

Q

Find the S-matrix of lossless quarter and half wavelength long traces.

- Lossless $\gamma = j\beta$
- $S_{11}=S_{11}=0$ since matched
- $V_2^- = V_1^+ \times e^{-j\beta l}$
- $S_{21} = \left. \frac{V_2^-}{V_1^+} \right|_{V_2^+=0} = e^{-j\beta l}$
- For quarter wave $e^{-j\beta l} = e^{-j\frac{2\pi\lambda}{\lambda} \frac{1}{4}} = e^{-j\frac{\pi}{2}} = -j \Rightarrow S_{21} = S_{12} = -j$
- So, there is a 100% transmission ($|S_{12}| = 1$) with 90 degree phase shift
- For half wave $e^{-j\beta l} = e^{-j\frac{2\pi\lambda}{\lambda} \frac{1}{2}} = e^{-j\pi} = -1 \Rightarrow S_{21} = A_{12} = -1$
- So, there is a 100% transmission ($|S_{12}| = 1$) with 180 degree phase shift



Matched reciprocal and lossless network

- A high-speed network can be any device or system with one or more ports
- Three important parameters that we look for
- **Matched:** $S_{ii}=0$ i.e. diagonal elements are zero
- **Reciprocal:** $[S] = [S]^t$ i.e. i and j are interchangeable in S_{ij}
- **Lossless:** $[S][S]^{*t} = [U]$,
 - The product of any column with the conjugate of the same column gives 1
 - The product of any column with the conjugate of any other column gives 0
- **Matched** means the ports are having the same impedance of characteristic impedance – no reflections back in to the inputs
- **Reciprocal:** Networks with no active devices, ferrites, or plasmas and are just made up of waveguide and transmission lines or antennas
- **Lossless** means no power is dissipated in the network

Q

- A two-port network is known to have the scattering matrix:
- $$[S] = \begin{bmatrix} 0.15 \angle 0^\circ & 0.85 \angle -45^\circ \\ 0.85 \angle 45^\circ & 0.2 \angle 0^\circ \end{bmatrix}$$
- Determine if the network is reciprocal and lossless.
 - If port 2 is terminated with a matched load, what is the return loss seen at port 1?

Because $[S]$ is **not symmetric**, the network is **not reciprocal**.

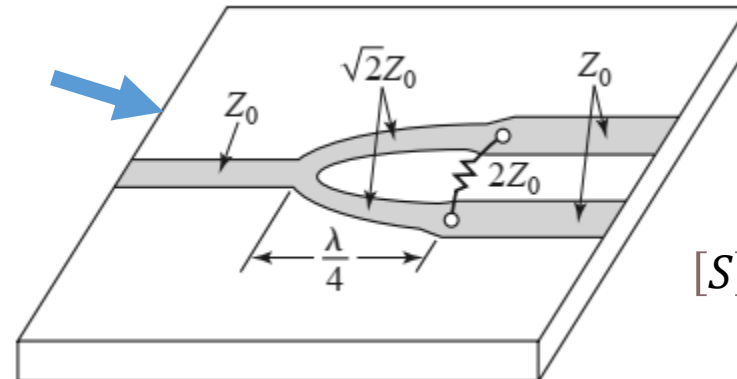
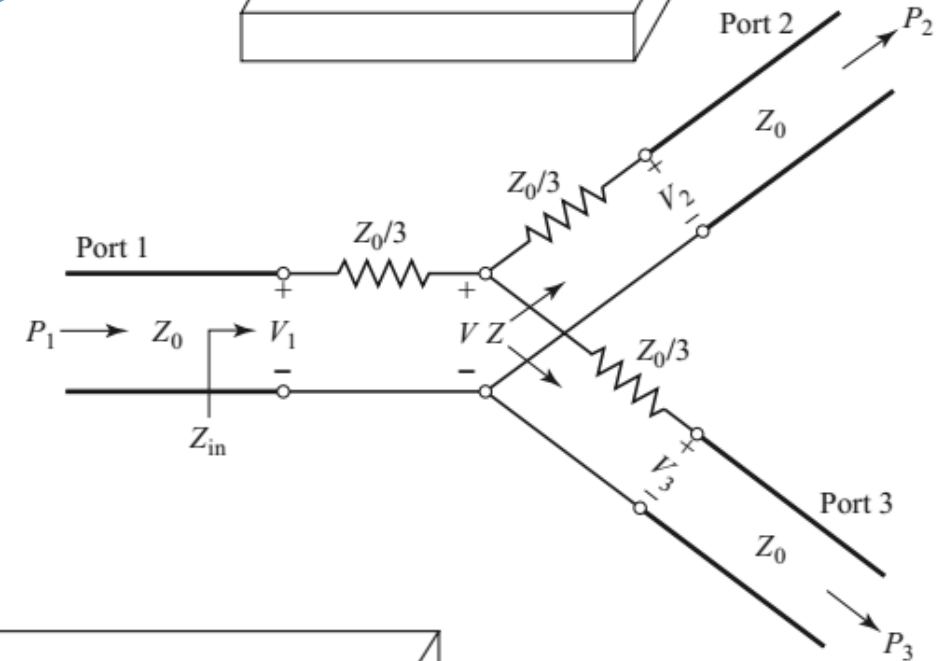
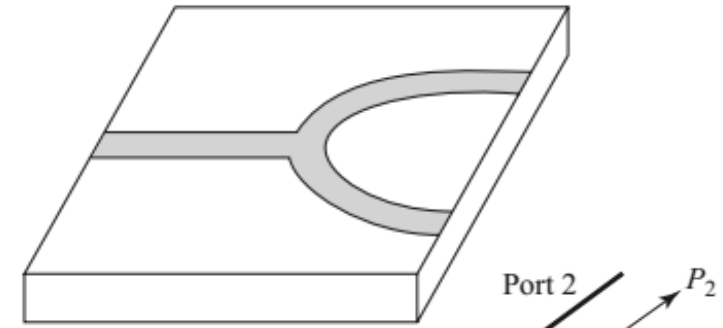
- **To be lossless, the product of any column with the conjugate of the same column gives unity,**
- Here, $|S_{11}|^2 + |S_{21}|^2 = (0.15)^2 + (0.85)^2 = 0.745 \neq 1$
- Hence, **not lossless**.

When port 2 is terminated with a matched load, the reflection coefficient seen at port 1 is $\Gamma = S_{11} = 0.15$.

So the return loss is $RL = -20 \log |\Gamma| = -20 \log(0.15) = 16.5 \text{ dB}$.

Common PCB structures and their behaviors

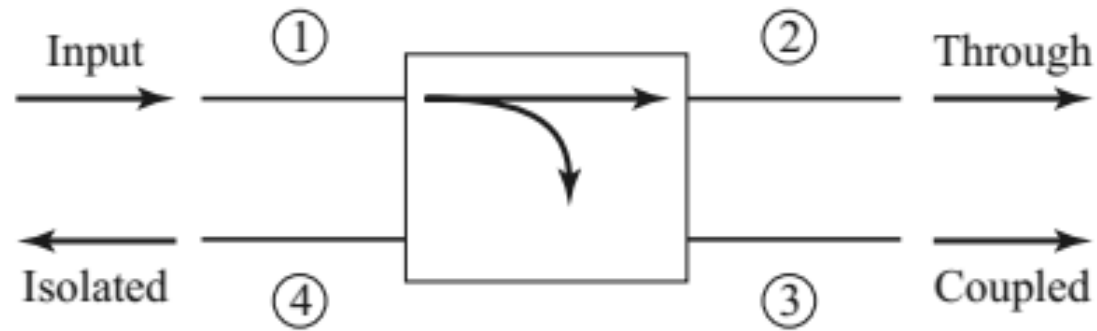
- It is **impossible** to construct a **three-port lossless** reciprocal network that is **matched** at all ports.
- So a simple trace that branches (called **T-junction divider**) **won't be matched**
- One way to achieve matching is to add resistors (**resistive power divider**) so that a **part of the power is dissipated to get matching**
- A better way will be to use a **Wilkinson power divider** that **will be matched**, but **won't dissipate power unless there is a mismatch between the outputs**



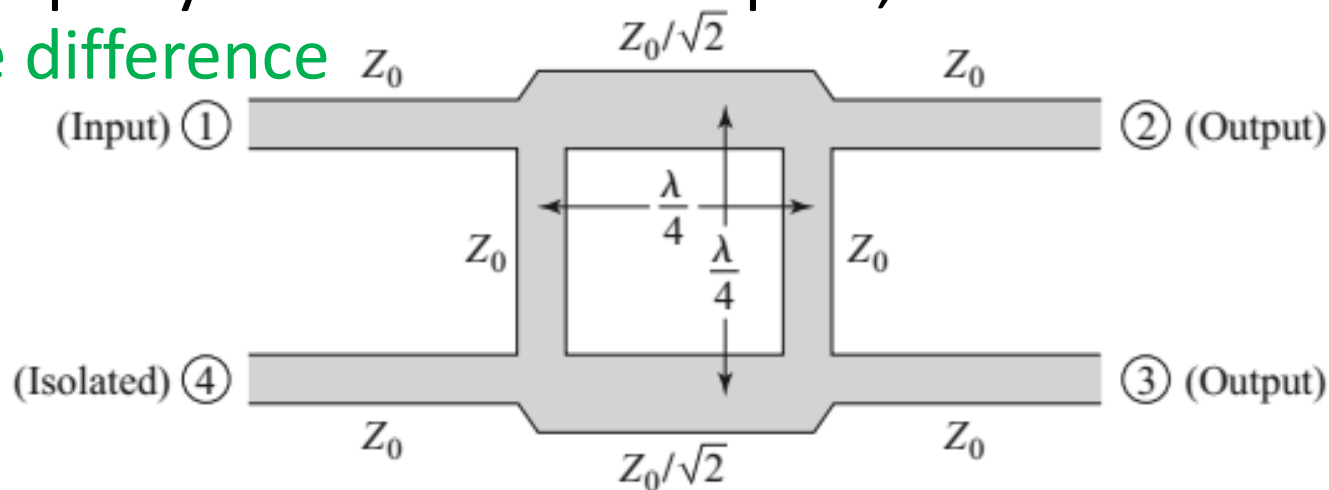
$$[S] = \frac{1}{\sqrt{2}} \begin{bmatrix} 0 & -j & -j \\ -j & 0 & 0 \\ -j & 0 & 0 \end{bmatrix}$$

Common PCB structures and their behaviors

- Any reciprocal, lossless, matched four-port network is a directional coupler
- Hence, a four-port network, if matched, reciprocal, and lossless, will have the following ports



- One common four-port power divider is a **quadrature hybrid** that divides the input power equally between two outputs, with the two outputs having a **90-degree phase difference**



$$[S] = \frac{1}{\sqrt{2}} \begin{bmatrix} 0 & 1 & j & 0 \\ 1 & 0 & 0 & j \\ j & 0 & 0 & 1 \\ 0 & j & 1 & 0 \end{bmatrix}$$

IPC-2221A lists four types of transmission lines

- Online calculators: <https://www.digikey.com/en/resources/conversion-calculators/conversion-calculator-pcb-trace-impedance>

